Hypothesis testing

1) $X\ B(20, 0.4)$. Calculate:

a) $P(X = 5)$

b) $P(X = 10)$

c) $P(X \leq 2)$

d) $P(X \geq 18)$

?

?

?

?

0.00000504

2) Wanda rolls a fair dice eight times.

a) Suggest a suitable model for the random variable X, the number of times the dice lands on five.

?

b) Calculate:

i. $P(X = 2)$

ii. $P(X \geq 4)$

?

?

Experimental

i.e. Dealing with collected data.

Chp1: Data

Methods of sampling, types of data, and populations vs samples.

Chp2: Measures of Location/Spread

Statistics used to summarise data, including mean, standard deviation, quartiles, percentiles. Use of linear interpolation for estimating medians/quartiles.

Chp3: Representation of Data

Producing and interpreting visual representations of data, including box plots and histograms.

Chp4: Correlation

Measuring how related two variables are, and using linear regression to predict values.

Theoretical

Deal with probabilities and modelling to make inferences about what we 'expect' to see or make predictions, often using this to reason about/contrast with experimentally collected data.

Chp5: Probability

Venn Diagrams, mutually exclusive + independent events, tree diagrams.

Chp6: Statistical Distributions

Common distributions used to easily find probabilities under certain modelling conditions, e.g. binomial distribution.

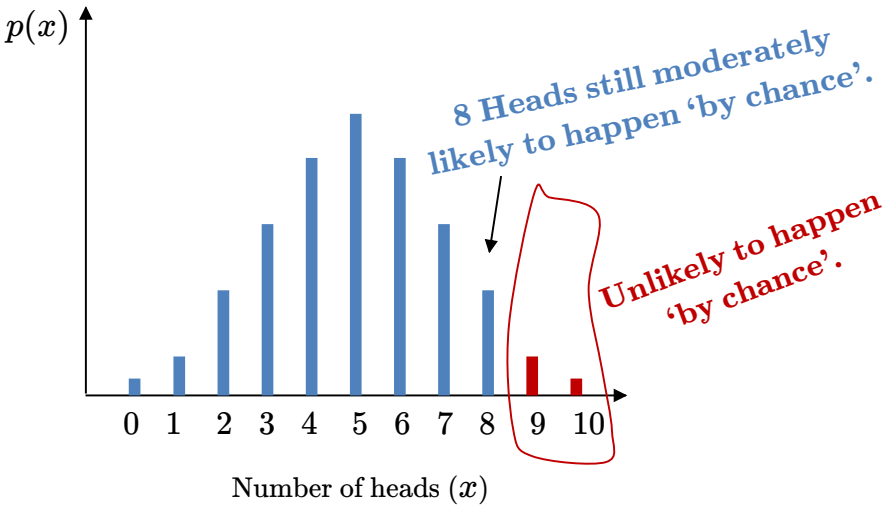
Chp7: Hypothesis Testing

Determining how likely observed data would have happened 'by chance', and making subsequent deductions.



I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

Our intuition is that the further away we are from the 'expected' number of heads (i.e. 5 heads out of 10), the more unlikely it is.





I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

A hypothesis is a statement made about the value of a **population parameter** that we wish to test by collecting evidence in the form of a sample.

! The **null hypothesis**, H_0 is the default position, i.e. that nothing has changed, unless proven otherwise.

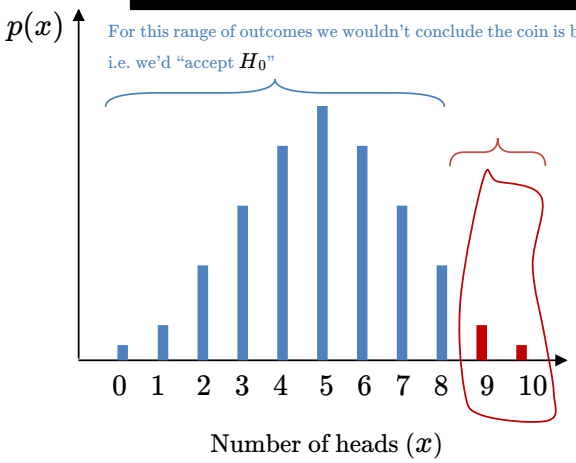
! The **alternative hypothesis**, H_1, is that there has been some change in the population parameter.

In this context...

?

?

?



For this range of outcomes we'd conclude that this number of heads was too unlikely to happen by chance, and hence reject H_0 (i.e. that coin was fair) and accept H_1 (i.e. that coin was biased).

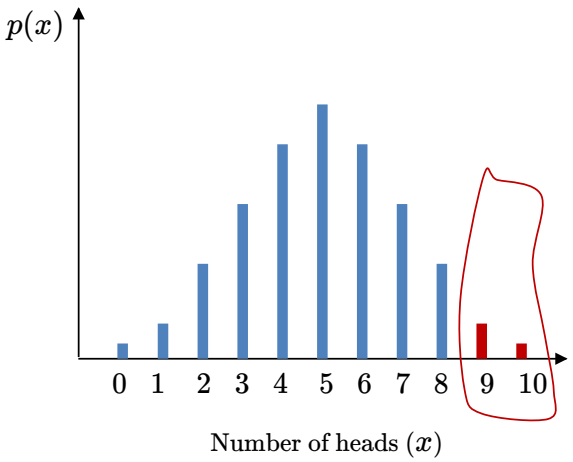


I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

In this context...

In a hypothesis test, the evidence from the sample is a **test statistic**.

?



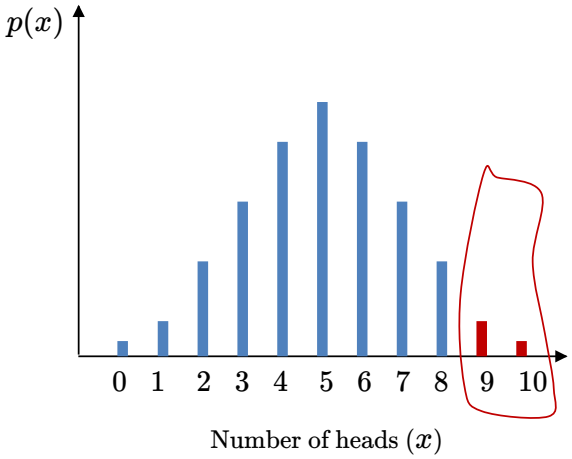


I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

In this context...

! The **level of significance** α is the maximum probability where we would reject the null hypothesis. This is usually 5% or 1%.

?



Hypothesis testing in a nutshell then is:

1. We have some hypothesis we wish to see if true (e.g. coin is biased towards heads), so...
2. We collect some sample data by throwing the coin (giving us our 'test statistic') and...
3. If that number of heads (or more) is sufficiently unlikely to have emerged 'just by chance', then we conclude that our (alternative) hypothesis is correct, i.e. the coin is biased.



[Textbook] John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost.

We said that our two hypotheses are about the population parameter.

Suppose p is the probability of a coin landing heads.

Null hypothesis: ?

Alternative hypothesis: ?

The latter is known as a '**one-tailed test**' because we're saying the coin is biased one way or the other (i.e. $p > 0.5$ or $p < 0.5$).
But we could also have had the hypothesis 'the coin is biased (either way)', i.e. $p \neq 0.5$. This is known as a **two-tailed test**.

[Textbook] An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. 3 people say they do.
a) Write down a suitable test statistic.
b) Write down two suitable hypotheses.
c) Explain the condition under which the null hypothesis would be rejected.

a ?

b ?

c ?

Test Your Understanding

In the UK, 5% of students turn up late to school each day. Miss Weir wishes to determine, to a 10% significance level if her school has a problem with attendance. She stands at the front gate one day and finds that 6 of the 40 students who pass her are late.

a) Write down a suitable test statistic.

b) Write down two suitable hypotheses.

c) Explain the condition under which the null hypothesis would be rejected.

a ?

b ?

c ?

Exercise 7A - Statistics 1 Pages 100-101

1 **a** Explain what you understand by a hypothesis test.

b Define a null hypothesis and an alternative hypothesis and state the symbols used for each.

c Define a test statistic.

a?

b?

c?

2 For each of these hypotheses, state whether the hypotheses given describe a one-tailed or a two-tailed test:

a $H_0: p = 0.8$, $H_1: p > 0.8$ **b** $H_0: p = 0.6$, $H_1: p \neq 0.6$ **c** $H_0: p = 0.2$, $H_1: p < 0.2$

a?

b?

c?

3 Dmitri wants to see whether a dice is biased towards the value 6. He throws the dice 60 times and counts the number of sixes he gets.

a Describe the test statistic.

b Write down a suitable null hypothesis to test this dice.

c Write down a suitable alternative hypothesis to test this dice.

a?

b?

c?

Shell wants to test to see whether a coin is biased. She tosses the coin 100 times and counts the number of times she gets a head. Shell says that her test statistic is the probability of the coin landing on heads.

a Explain the mistake that Shell has made and state the correct test statistic for her test.

b Write down a suitable null hypothesis to test this coin.

c Write down a suitable alternative hypothesis to test this coin.

a?

b?

c?

5 In a manufacturing process the proportion (p) of faulty articles has been found, from long experience, to be 0.1.
A sample of 100 articles from a new manufacturing process is tested, and 8 are found to be faulty.
The manufacturers wish to test at the 5% level of significance whether or not there has been a reduction in the proportion of faulty articles.

a Suggest a suitable test statistic.

b Write down two suitable hypotheses.

c Explain the condition under which the null hypothesis is rejected.

a?

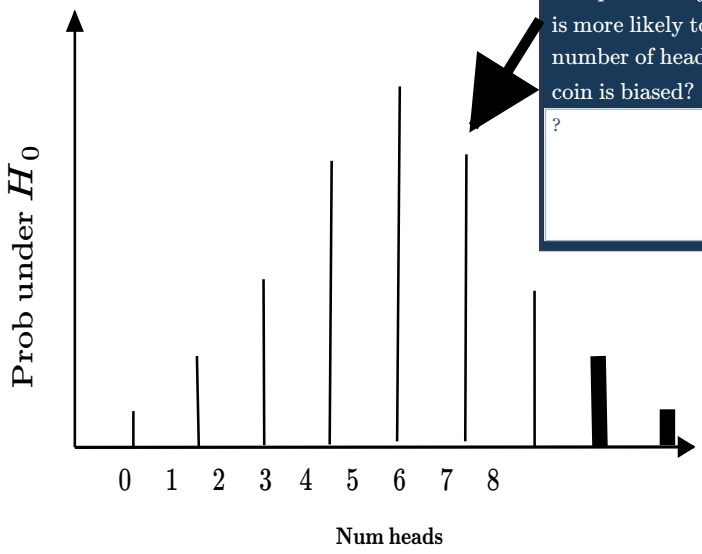
b?

c?

Critical Regions and Values 16

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected?**

As before, we're interested how likely a given outcome is likely to happen 'just by chance' under the null hypothesis (i.e. when the coin is not biased).



The probability of getting exactly 5 heads is only 22%, which is more likely to not happen than to happen. If we saw this number of heads, why would it not be sensible to think the coin is biased?

?

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

What's the probability that we would see **6 heads**, or an **even more extreme value**? Is this sufficiently unlikely to support John's claim that the coin is biased?

?

What's the probability that we would see **7 heads**, or an **even more extreme value**?

?

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

The **critical region** is the range of values of the test statistic that would lead to you rejecting H_0

?

The value(s) on the boundary of the critical region are called **critical value(s)**.

Critical value:

?

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Quickfire Critical Regions 19

Determine the critical region when we throw a coin where we're trying to establish if there's the specified bias, given the specified number of throws, when the level of significance is 5%.

Coin thrown 5 times. Trying to establish if biased towards heads.

$p = 0.5, n = 5$

x	$P(X \leq x)$
0	0.0312
1	0.1875
2	0.5000
3	0.8125
4	0.9688

Critical region:

?

Coin thrown 10 times. Trying to establish if biased towards heads.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

?

Coin thrown 10 times. Trying to establish if biased towards tails.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

?

Actual Significance Level 20

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

We saw earlier that the critical region was $X \geq 7$, i.e. the region in which John would reject the null hypothesis (and conclude the coin was biased).

We ensured that $P(X \geq 7)$ was less than the significance level of 5%.
But what actually is $P(X \geq 7)$?

$\mathbf{P(X \geq 7)} =$? **?**

C.D.F. Binomial table:
$p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Two-tailed test

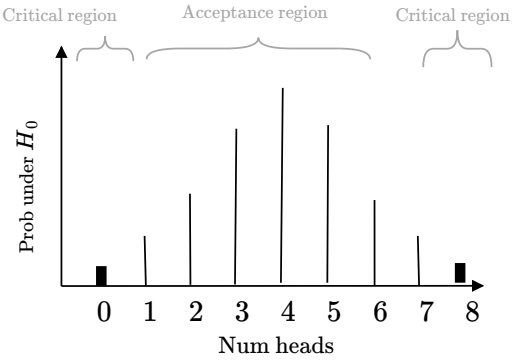
Suppose I threw a coin 8 times and was now interested in how may heads would suggest it was a **biased coin** (i.e. either way!). How do we work out the critical values now, with 5% significance?

We split the 5% so there's 2.5% at

?

?

?



C.D.F. Binomial table: $p = 0.5, n = 8$	
x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
...	...
6	0.9648
7	0.9961
8	1

Test Your Understanding

A random variable X has binomial distribution $B(40, p)$. A single observation is used to test $H_0 : p = 0.25$ against $H_1 : p \neq 0.25$.

The $\neq$ indicates bias either way, i.e. two-tailed.

a) Using the 2% level of significance, find the critical region of this test. The probability in each tail should be as close as possible to 0.01.

b) Write down the actual significance level of the test.

This means you find the closest to 0.01 (even if slightly above) rather than the closest under 0.01

a ?

b ?

C.D.F. Binomial table: $p = 0.25, n = 40$	
x	$P(X \leq x)$
2	0.0010
3	0.0047
4	0.0160
5	0.0433
16	0.9884
17	0.9953
18	0.9983
19	0.9994

Warning: Textbook has several typos in this example.

2 A test statistic has a distribution B(10, p). Given that H_0: $p = 0.2$, H_1: $p > 0.2$, find the critical region for the test using a 5% significance level.

?

3 A random variable has a distribution B(20, p). A single observation is used to test H_0: $p = 0.15$ against H_1: $p < 0.15$. Using a 5% level of significance, find the critical region of this test.

?

4 A random variable has distribution B(20, p). A single observation is used to test H_0: $p = 0.4$ against H_1: $p \neq 0.4$.

a Using the 5% level of significance, find the critical region of this test. **(3 marks)**

b Write down the actual significance level of the test. **(1 mark)**

a?

b?

5 A test statistic has a distribution B(20, p). Given that H_0: $p = 0.18$, H_1: $p < 0.18$, find the critical region for the test using a 1% level of significance.

?

6 A random variable has distribution B(10, p). A single observation is used to test H_0: $p = 0.22$ against H_1: $p \neq 0.22$.

a Using a 1% level of significance, find the critical region of this test. The probability in each tail should be as close as possible to 0.005. **(3 marks)**

b Write down the actual significance level of the test. **(2 marks)**

a ?

b ?

7 A mechanical component fails, on average, 3 times out of every 10. An engineer designs a new system of manufacture that he believes reduces the likelihood of failure. He tests a sample of 20 components made using his new system.

a Describe the test statistic. **(1 mark)**

b State suitable null and alternative hypotheses. **(2 marks)**

c Using a 5% level of significance, find the critical region for a test to check his belief, ensuring the probability is as close as possible to 0.05. **(3 marks)**

d Write down the actual significance level of the test. **(1 mark)**

a?

b?

c?

d?

8 Seedlings come in trays of 36. On average, 12 seedlings survive to be planted on. A gardener decides to use a new fertiliser on the seedlings which she believes will improve the number that survive.

a Describe the test statistic and state suitable null and alternative hypotheses. **(3 marks)**

b Using a 10% level of significance, find the critical region for a test to check her belief. **(3 marks)**

c State the probability of incorrectly rejecting H_0 using this critical region. **(1 mark)**

a?

b?

c?

9 A restaurant owner notices that her customers typically choose lasagne one fifth of the time. She changes the recipe and believes this will change the proportion of customers choosing lasagne.

a Suggest a model and state suitable null and alternative hypotheses. **(3 marks)**

She takes a random sample of 25 customers.

b Find, at the 5% level of significance, the critical region for a test to check her belief. **(4 marks)**

c State the probability of incorrectly rejecting H_0. **(1 mark)**

a?

b?

c?

Doing a full one-tailed hypothesis test

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

?

?

?

?

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

STEP 2: Write null and alternative hypotheses.

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.
i.e. Determine probability we'd see this outcome just by chance.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

Alternative method using critical regions

We can also find the critical region and see if the test statistic lies within it.

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

X is number of heads.
p is probability of heads.

$X \ B\,(8, p)$

$H_0 : p = 0.5$
$H_1 : p > 0.5$

?

?

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

STEP 2: Write null and alternative hypotheses.

STEP 3 (Alternative): Determine critical region.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

More on p-values

Sheila wants to know if a coin is biased towards heads and throws it a large number of times, counting the number of heads. The p-value is less than 0.03.
Conduct a hypothesis test at the 5% significance level.

?

Ordinarily we'd calculate the probability of seeing the observed number of heads 'or more extreme'. But this has already been done for us (i.e. the p-value), so we just need to compare this against the threshold.

Further Example

[Textbook] The standard treatment for a particular disease has a $\frac{2}{5}$ probability of success. A certain doctor has undertaken research in this area and has produced a new drug which has been successful with 11 out of 20 patients. The doctor claims the new drug represents an improvement on the standard treatment. Test, at the 5% significance level, the claim made by the doctor.



?

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

?

STEP 2: Write null and alternative hypotheses.

?

0.1275

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.

?

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

Edexcel S2 Jan 2011 Q2

A student takes a multiple choice test. The test is made up of 10 questions each with 5 possible answers. The student gets 4 questions correct. Her teacher claims she was guessing the answers. Using a one tailed test, at the 5% level of significance, test whether or not there is evidence to reject the teacher's claim.

State your hypotheses clearly. **(6)**

?

Jessica complains that a dice she is using is biased so that she is less likely to get a 6. She decides to test at the 5% significance level the hypothesis that the dice is biased against a six.

(i) Write down the null and alternative hypothesis that Jessica will use.

Jessica throws the dice 20 times and gets just one six.

(ii) Find the *p*-value for this test.

(iii) What is Jessica's conclusion?

(i) ?

(ii) ?

(iii) ?

Hassan complains that a dice he is using is biased so that he is more likely to get a 1 than any other number. He decides to test at the 10% significance level the hypothesis that the dice is biased towards a 1.
(i) Write down the null and alternative hypothesis that Hassan will use.
Hassan throws the dice 12 times and gets 5 ones.
(ii) Find the *p*-value for this test.
(iii)What is Hassan's conclusion?

(i) ?

(ii) ?

(ii) ?

A bus company claims that a particular service is on time in 90% of journeys. I think it is less than this. I decide to test this at the 5% significance level.
(i) Write down the null and alternative hypothesis that I should use.
In my next 15 journeys, the bus is on time 10 times.
(ii) Find the *p*-value for this test.
(iii)What is my conclusion?

(i) ?

(ii) ?

(iii ?

Over a long period of time, it is found that a particular bus service is on time in 80% of journeys. The bus company claims that it has made improvements to the service and that the bus is now on time more often. I decide to test this at the 10% significance level.

(i) Write down the null and alternative hypotheses that I should use.

In my next 16 journeys, the bus is on time 15 times.

(ii) Find the p-value for this hypothesis test.

(iii)What is my conclusion?

(i) ?

(ii) ?

(iii) ?

Exercise 7C - Statistics 1 Pages 106-107 38

1 A single observation, x, is taken from a binomial distribution B(10, p) and a value of 5 is obtained. Use this observation to test H_0: $p = 0.25$ against H_1: $p > 0.25$ using a 5% significance level.

?

2 A random variable has distribution $X \sim B(10, p)$. A single observation of $x = 1$ is taken from this distribution. Test, at the 5% significance level, H_0: $p = 0.4$ against H_1: $p < 0.4$.

?

3 A single observation, x, is taken from a binomial distribution B(20, p) and a value of 10 is obtained. Use this observation to test H_0: $p = 0.3$ against H_1: $p > 0.3$ using a 5% significance level.

Distribution, B(20, 0.30)

?

4 A random variable has distribution $X \sim B(20, p)$. A single observation of $x = 3$ is taken from this distribution. Test, at the 1% significance level, H_0: $p = 0.45$ against H_1: $p < 0.45$.

?

5 A single observation, x, is taken from a binomial distribution B(20, p) and a value of 2 is obtained. Use this observation to test H_0: $p = 0.28$ against H_1: $p < 0.28$ using a 5% significance level.

?

6 A random variable has distribution $X \sim B(8, p)$. A single observation of $x = 7$ is taken from this distribution. Test, at the 5% significance level, H_0: $p = 0.32$ against H_1: $p > 0.32$.

?

7 A dice used in playing a board game is suspected of not giving the number 6 often enough. During a particular game it was rolled 12 times and only one 6 appeared. Does this represent significant evidence, at the 5% level of significance, that the probability of a 6 on this dice is less than $\frac{1}{6}$?

?

8 The success rate of the standard treatment for patients suffering from a particular skin disease is claimed to be 68%.

a In a sample of n patients, X is the number for which the treatment is successful. Write down a suitable distribution to model X. Give reasons for your choice of model.

A random sample of 10 patients receives the standard treatment and in only 3 cases was the treatment successful. It is thought that the standard treatment was not as effective as it is claimed.

b Test the claim at the 5% level of significance.

a?

b?

We have already seen that if we're interest in bias 'either way', we have two tails, and therefore have to split the critical region by **halving the significance level at each end**.

Over a long period of time it has been found that in Enrico's restaurant the ratio of non-veg to veg meals is 2 to 1. In Manuel's restaurant in a random sample of 10 people ordering meals, 1 ordered a vegetarian meal. Using a 5% level of significance, test whether or not the proportion of people eating veg meals in Manuel's restaurant is different to that in Enrico's restaurant.

?

Edexcel S2 Jan 2006 Q7a

A teacher thinks that 20% of the pupils in a school read the Deano comic regularly.

He chooses 20 pupils at random and finds 9 of them read the Deano.

(*a*) (i) Test, at the 5% level of significance, whether or not there is evidence that the percentage of pupils that read the Deano is different from 20%. State your hypotheses clearly.

(ii) State all the possible numbers of pupils that read the Deano from a sample of size 20 that will make the test in part (*a*)(i) significant at the 5% level. **(9)**

(a)(i) ?

(ii) ?

1 A single observation, x, is taken from a binomial distribution B(30, p) and a value of 10 is obtained. Use this observation to test H_0: $p = 0.5$ against H_1: $p \neq 0.5$ using a 5% significance level.

?

2 A random variable has distribution $X \sim B(25, p)$. A single observation of $x = 10$ is taken from this distribution. Test, at the 10% significance level, H_0: $p = 0.3$ against H_1: $p \neq 0.3$.

?

3 A single observation, x, is taken from a binomial distribution B(10, p) and a value of 9 is obtained. Use this observation to test H_0: $p = 0.75$ against H_1: $p \neq 0.75$ using a 5% significance level.

?

4 A random variable has distribution $X \sim B(20, p)$. A single observation of $x = 1$ is taken from this distribution. Test, at the 1% significance level, H_0: $p = 0.6$ against H_1: $p \neq 0.6$.

?

5 A random variable has distribution $X \sim B(50, p)$. A single observation of $x = 4$ is taken from this distribution. Test, at the 2% significance level, $H_0: p = 0.02$ against $H_1: p \neq 0.02$.

?

6 A coin is tossed 20 times, and lands on heads 6 times. Use a two-tailed test with a 5% significance level to determine whether there is sufficient evidence to conclude that the coin is biased.

?

7 The national proportion of people experiencing complications after having a particular operation in hospitals is 20%. A hospital decides to take a sample of size 20 from their records.

a Find critical regions, at the 5% level of significance, to test whether or not their proportion of complications differs from the national proportion. The probability in each tail should be as close to 2.5% as possible. **(5 marks)**

b State the actual significance level of the test. **(1 mark)**

The hospital finds that 8 of their 20 patients experienced complications.

c Comment on this finding in light of your critical regions. **(2 marks)**

a?

b?

c?

8 A machine makes glass bowls and it is observed that one in ten of the bowls have hairline cracks in them. The production process is modified and a sample of 20 bowls is taken. 1 of the bowls is cracked. Test, at the 10% level of significance, the hypothesis that the proportion of cracked bowls has changed as a result of the change in the production process. State your hypotheses clearly. **(7 marks)**

a?

b?

9 Over a period of time, Agnetha has discovered that the carrots that she grows have a 25% chance of being longer than 7 cm. She tries a new type of fertiliser. In a random sample of 30 carrots, 13 are longer than 7 cm. Agnetha claims that the new fertiliser has changed the probability of a carrot being longer than 7 cm. Test Agnetha's claim at the 5% significance level. State your hypotheses clearly. **(7 marks)**

?

10 A standard blood test is able to diagnose a particular disease with probability 0.96.
A manufacturer suggests that a cheaper test will have the same probability of success.
It conducts a clinical trial on 75 patients. The new test correctly diagnoses 63 of these patients.
Test the manufacturer's claim at the 10% level, stating your hypotheses clearly. **(7 marks)**

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